Topic 4: Analog to Digital Conversion

CSE 4405: Data and Telecommunications

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Learning Objectives

1. Explain why analog signals are digitized before transport, storage, and processing
2. Apply sampling intuition and the Nyquist condition without memorizing isolated formulas
3. Explain uniform and non-uniform quantization, quantization error, and quality/bit-rate tradeoffs
4. Derive and use the PCM bit-rate relation $R_b = f_s \times n_b$
5. Distinguish PCM, DPCM, ADPCM, and Delta Modulation through system intuition and diagrams
6. Connect analog-to-digital conversion to telephony, multimedia compression, and Topic 5 digital modulation

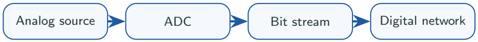
Reading focus: Forouzan 5e, Ch. 4 (digital transmission, analog-to-digital conversion sequence); Topic 2 Ch. 3 pp. 81–83 for Nyquist/Shannon background; Goldsmith Ch. 5 pp. 126–132 as a bridge to digital passband systems.

Why Convert Analog Sources into Digital Bits?

Core reasoning

Digital representation gives repeatability and control: noise can be regenerated away, processing is programmable, and transport/storage become network-friendly.

- **Noise resilience:** repeaters regenerate symbols better than analog amplification.
- **System integration:** switching, multiplexing, encryption, and compression all operate naturally on bits.
- **Scalability:** one packetized backbone can carry voice, video, and sensor streams.



Interpretation: ADC is the gateway from physical waveforms to bit-oriented communication.

Analog-to-Digital Conversion

Motivation

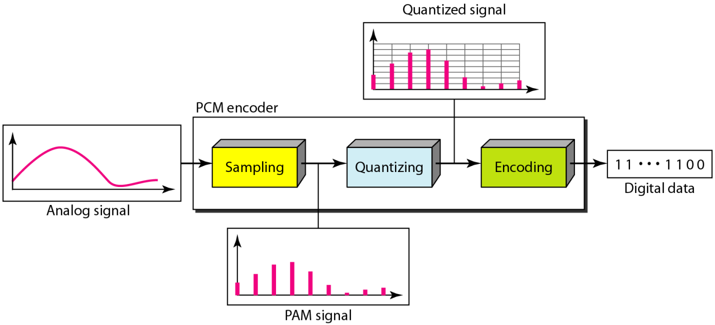
Digital representations are easier to regenerate, store, switch, and process reliably than analog waveforms.

- **PCM (Pulse Code Modulation)**: sample, quantize, and encode amplitude values
- **DPCM/ADPCM**: encode prediction error, optionally with adaptive step behavior
- **DM (Delta Modulation)**: encode change between successive samples

Topic order here: PCM first, then DPCM/ADPCM refinements, then DM.

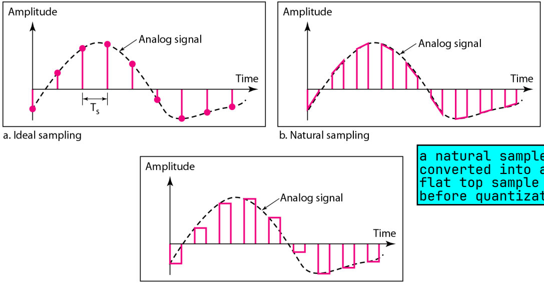
PCM Encoder Structure

Figure: Components of PCM encoder



- Sampling measures analog amplitude at uniform time intervals.
- Pulse amplitude modulation is an intermediate concept, not the final communication format.
- In PCM, sampled values become the input to quantization.

Figure: Sampling methods for PCM



a. Ideal sampling

b. Natural sampling

c. Flat-top sampling

a natural sample is converted into a flat top sample before quantization.

Nyquist Sampling Condition

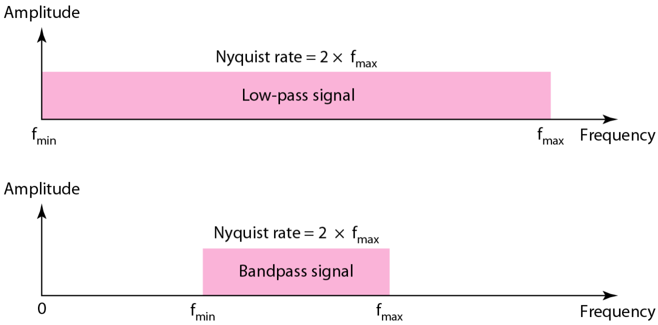
$$f_s \geq 2f_{\max}$$

- f_s: sampling rate
- $f_{\max}$: highest frequency present in the source signal

- Sampling below this limit causes aliasing.
- At the ideal threshold, or above it, reconstruction is possible only under band-limited assumptions and proper filtering.
- Telephony example: $f_{\max} = 4$ kHz $\Rightarrow f_s = 8$ kHz.

Nyquist Condition: Visual Interpretation

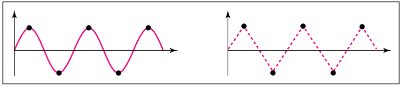
Figure: Nyquist sampling for low-pass and bandpass signals



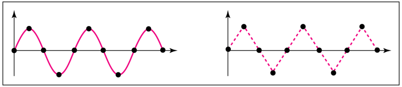
Example 4.1: Intuition with a Sine Wave

- Compare $f_s = 4f$, $f_s = 2f$, and $f_s = f$.
- $f_s = 2f$ is the ideal minimum boundary for a pure tone, not a robust design target.
- Oversampling gives practical guard margin for filtering, timing, and noise.
- Undersampling ($f_s < f_{\text{Nyquist}}$) distorts recovered shape.

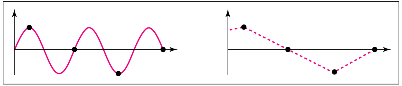
Figure: Recovery at different sampling rates



a. Nyquist rate sampling: $f_s = 2f$



b. Oversampling: $f_s = 4f$



c. Undersampling: $f_s = f$

Example 4.2: Telephony Sampling Rate

Voice digitization assumption

Telephone voice is band-limited to approximately 0 to 4000 Hz.

$$f_s = 2 \times 4000 = 8000 \text{ samples/s}$$

Result: standard PCM telephony uses 8 kHz sampling.

Quantization

So when bits increase, L increases, and Δ becomes smaller. A smaller step size means each sample is rounded by a smaller amount, so the quantization error decreases.

By ideal conditions, the formula SQNR_dB ≈ 6.02n_b + 1.76 assumes a simplified "perfect ADC" situation:
--> The input is a full-scale sine wave, meaning it uses almost the entire allowed voltage range without clipping.
--> The quantizer is uniform, meaning all quantization levels are equally spaced.
--> The only noise is quantization noise. It ignores real-world noise such as thermal noise, circuit noise, clock jitter, distortion, and imperfect ADC hardware.
--> The signal is properly sampled, so there is no aliasing.

- Divide analog amplitude range $[V_{\text{min}}, V_{\text{max}}]$ into L zones.
- Zone height:

$$\Delta = \frac{V_{\text{max}} - V_{\text{min}}}{L}$$

- Replace each sample by nearest quantized level (midpoint representation).
- Quantization introduces error; increasing bits/sample reduces it.

$$\mathrm{SQNR}_{dB} \approx 6.02\, n_b + 1.76$$

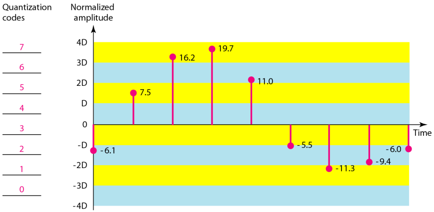
for ideal uniform PCM quantization of a full-scale sinusoidal input, where n_b is bits per sample.

Uniform vs Non-Uniform Quantization

	Uniform	**Non-uniform / companded**
Step size	Same Δ everywhere	Smaller near zero, larger at high amplitudes
Strength	Simple hardware and analysis	Better perceived speech quality at fixed bit depth
Cost	Can waste levels on rare large amplitudes	Compressor and expander must match
Examples	Linear PCM	μ-law and A-law PCM voice coding

Design idea: use more precision where the signal is more likely or more perceptually sensitive.

Figure: Quantization and encoding of sampled signal



Normalized PAM values	-1.22	1.50	3.24	3.94	2.20	-1.10	-2.26	-1.88	-1.20
Normalized quantized values	-1.50	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized error	-0.38	0	+0.26	-0.44	+0.30	-0.40	-0.24	+0.38	-0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

Example 4.3

Req.: $\mathrm{SQNR}_{dB} > 40$.

$$n_b \geq \frac{40 - 1.76}{6.02} \approx 6.35$$

Min.: $\boxed{\text{7 bits/sample}}$.

Telephony: 7–8 bits/sample.

PCM Encoding and Example 4.4

- After quantization, each sample level is encoded into an n_b-bit word.
- Number of quantization levels: $L = 2^{n_b}$, so $n_b = \log_2 L$.
- PCM bit-rate formula:

$$R_b = f_s \times n_b$$

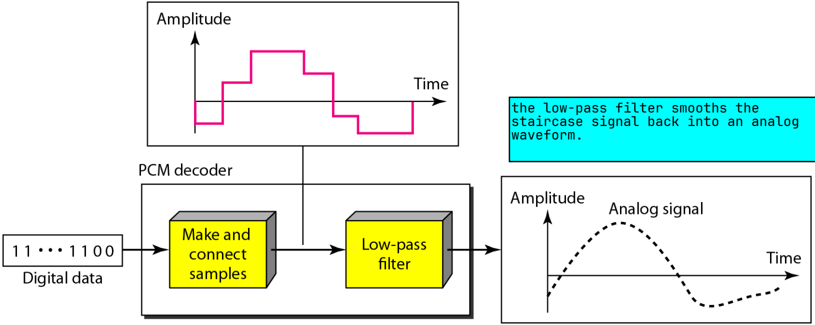
Example 4.4

Human voice: $f_{\text{max}} = 4$ kHz $\Rightarrow f_s = 8$ kHz.
If $n_b = 8$, then

$$R_b = 8000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

PCM Decoder and Signal Recovery

Figure: Components of PCM decoder



the low-pass filter smooths the staircase signal back into an analog waveform.

PCM Bandwidth and Example 4.5

$$B_{\text{min}} = c \times N \times \frac{1}{r}, \quad N = n_b f_s, \quad f_s \approx 2B_{\text{analog}}$$

For NRZ-like signaling ($r = 1$, average $c = \frac{1}{2}$):

$$B_{\text{min}} \approx n_b \, B_{\text{analog}}$$

Example 4.5

B_analog is the bandwidth of the original analog signal
B_min is the minimum required bandwidth for the PCM

If $B_{\text{analog}} = 4$ kHz and $n_b = 8$:

$$B_{\text{min}} \approx 8 \times 4 \text{ kHz} = 32 \text{ kHz}$$

Digitization improves robustness but increases required channel bandwidth.

Why Go Beyond PCM?

Observation

Adjacent speech samples are often similar. Sending full absolute amplitude every time can be redundant.

Sample sequence

$100,\ 102,\ 101,\ 103,\ 104, \ldots$

Difference sequence

$+2,\ -1,\ +2,\ +1, \ldots$

Idea behind DPCM/ADPCM

Differences are usually smaller than absolute values.
Smaller dynamic range can be quantized with fewer bits for similar perceived quality.

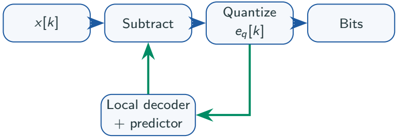
DPCM: Encode Prediction Error Instead of Raw Sample

Let $\hat{x}[k]$ be predicted sample from previous reconstructed values.

$$e[k] = x[k] - \hat{x}[k]$$

DPCM transmits quantized error $e_q[k]$, not $x[k]$. The receiver reconstructs $x_q[k] = \hat{x}[k] + e_q[k]$.

- Works well when signal changes smoothly
- A simple predictor is enough here (often previous-sample based)
- Receiver uses the same predictor and adds decoded error back



Example 4.6: DPCM with a Previous-Sample Predictor

Setup

Given $x_q[1] = 100$, use the previous reconstructed sample as predictor:

$$\hat{x}[k] = x_q[k-1]$$

Assume these small integer errors are reproduced exactly.

$$e[2] = 102 - 100 = +2$$
$$e[3] = 101 - 102 = -1$$
$$e[4] = 103 - 101 = +2$$

k	$x[k]$	$\hat{x}[k]$	$e[k]$
2	102	100	$+2$
3	101	102	-1
4	103	101	$+2$

Takeaway

DPCM sends small corrections $+2, -1, +2$, so the error range can often be quantized with fewer bits.

ADPCM: Make DPCM Adaptive

ADPCM extension

Signal behavior changes over time. A fixed quantizer step is inefficient. ADPCM adapts step size and/or predictor parameters as the signal evolves.

Method	Rate
PCM	64 kbps
DPCM	32–48 kbps
ADPCM	16–40 kbps

Typical voice operating points; exact values vary by standard and quality target.

- Fast-changing regions: larger step to avoid overload
- Slow-changing regions: smaller step to reduce granular error
- Better quality-rate tradeoff than fixed DPCM

Example 4.7: ADPCM Adapts to Signal Changes

Sample behavior

Use the same previous-sample predictor, and suppose the source evolves as

$$40,\ 41,\ 42,\ 50,\ 58,\ 59$$

Small changes are followed by a sudden rise, then the signal settles again.

- Early samples need only a small step size: $\Delta[k] = 1$
- Large positive errors trigger a larger step: $\Delta[k] = 4$
- When the slope becomes gentle again, the step shrinks back

k	$x[k]$	$\hat{x}[k]$	$e[k]$	$\Delta[k]$
2	41	40	$+1$	1
3	42	41	$+1$	1
4	50	42	$+8$	4
5	58	50	$+8$	4
6	59	58	$+1$	1

Takeaway

With a fixed small step, the $+8$ errors would be poorly represented.
Here $\Delta[k]$ shows step-size intuition, not a full ADPCM codebook.

PCM vs DPCM vs ADPCM at a Glance

Method	What is coded	Complexity	Strength	Main limitation
PCM	Absolute sample level	Low	Simple and robust baseline	Higher bit rate for given quality
DPCM	Prediction error	Medium	Exploits sample correlation	Performance depends on predictor and fixed step choice
ADPCM	Adaptively quantized prediction error	Medium-high	Better quality-rate tradeoff for speech/audio	More control logic and state synchronization

Pattern

Each step from PCM to ADPCM spends more algorithmic complexity to save transmission bits.

Delta Modulation (DM)

- DM reduces PCM complexity by encoding **change**, not absolute amplitude.
- Each bit indicates step up or step down from previous estimate.
- Simpler hardware, but slope-overload and granular-noise tradeoffs exist.

Figure: Delta modulation process

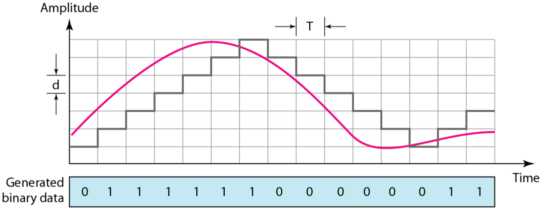


Figure: Delta modulation components

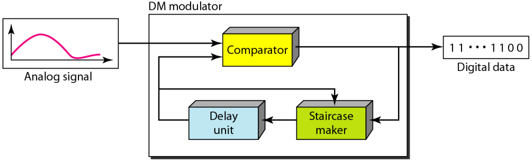
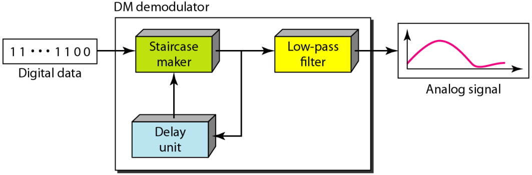


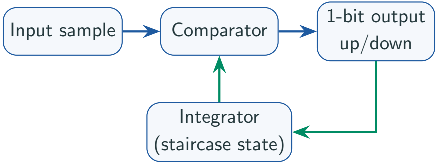
Figure: Delta demodulation components



Delta Modulation: 1-Bit Difference Signaling

Core idea

Delta Modulation (DM) is an extreme DPCM-like approach: transmit only one bit per sample to indicate **up** or **down**.



Advantage: simple hardware and one bit/sample. **Challenge:** very sensitive to step-size choice.

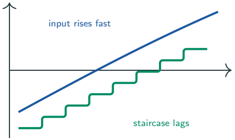
Two Classical DM Errors

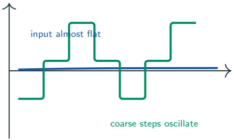
Slope overload

If input changes faster than staircase can climb/fall, the reconstruction lags badly.
Cause: step δ too small for steep regions.

Granular noise

If input is slowly varying but step δ is large, staircase oscillates around true value.
Cause: step too large for flat regions.





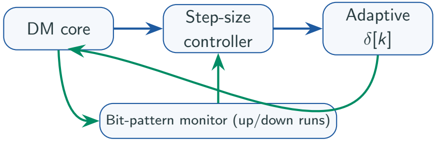
Design tension: small δ helps flat regions; large δ helps steep regions.

Adaptive Delta Modulation (ADM)

Fix

ADM changes step size dynamically: small when signal is calm, large when slope is steep.

- Several consecutive bits in same direction $\Rightarrow$ increase δ
- Alternating up/down bits $\Rightarrow$ decrease δ
- Reduces both slope overload and granular noise in practice



What to remember: ADM keeps DM's 1-bit idea but improves step-size management.

Where These Methods Appear in Practice

Method	Typical role	Why used
PCM	Baseline telephony and professional digital audio chains	Predictable quality, simple interoperability, easy switching/multiplexing
DPCM, ADPCM	Speech codecs and bandwidth-limited voice links	Good quality at lower bit rates than plain PCM
DM/ADM	Simple low-complexity systems and historical/teaching relevance	Very low encoder complexity and 1-bit conceptual clarity

System thinking: real designs choose among these using quality, available bit rate, complexity, and latency.

How ADC Enables Multimedia Compression

Pipeline logic

Compression works on digital representations, so analog capture must be digitized first.

- Compression exploits digital redundancy after ADC
- Error-control and packet transport are practical in this bit domain

Common Confusions and Quick Fixes

Common confusion	**Correct interpretation**
"Sampling already makes the signal digital."	Sampling makes time discrete, but amplitude is still continuous until quantization.
"Nyquist says exactly $f_s = 2f_{\max}$ always."	$2f_{\max}$ is the minimum ideal threshold; practical systems often sample above it.
"More quantization levels are always better."	Better quality, yes, but also higher bit rate and system cost. Design is a tradeoff.
"PCM and DM are unrelated."	DM can be viewed as an extreme 1-bit differential strategy in the same ADC design family.

Summary

- Analog-to-digital conversion is the gateway from physical analog sources to digital communication systems
- Sampling chooses **when** values are captured; Nyquist condition prevents aliasing
- Quantization chooses **how finely** amplitudes are represented; this introduces quantization error
- PCM combines sampling, quantization, and binary encoding, with $R_b = f_s \times n_b$
- DPCM/ADPCM improve efficiency by coding prediction error at lower rates
- Delta Modulation uses 1-bit up/down updates; ADM improves step-size behavior
- These ideas directly prepare Topic 5 digital modulation

References

1. B. A. Forouzan, *Data Communications and Networking*, 5th ed., Ch. 4, digital transmission: analog-to-digital conversion sequence (sampling, quantization, PCM, and delta modulation).
2. Topic 2 deck source base (Forouzan Ch. 3, pp. 81–83): Nyquist and Shannon background used here for sampling-rate intuition.
3. A. Goldsmith, *Wireless Communications*, Ch. 5, pp. 126–132 for passband/digital-system bridge context (why bit-oriented pipelines matter before digital modulation).

Page ranges can shift slightly across local/international printings

Terminology I

Term	Meaning in this topic
Sampling period (T_s)	Time interval between two consecutive samples
Sampling frequency (f_s)	Number of samples per second; $f_s = 1/T_s$
Nyquist rate	Minimum ideal low-pass sampling rate $2f_{\max}$ under band-limited reconstruction assumptions
Aliasing	Spectral overlap after undersampling that creates false low-frequency components
Anti-aliasing filter	Input low-pass filter before sampling to enforce band limitation
Quantization level	One allowed amplitude value used to represent a sample
Quantization step (Δ)	Spacing between adjacent quantization levels in uniform quantizers
Quantization error	Difference between true sample amplitude and chosen quantized level
SQNR	Ratio of signal power to quantization-noise power (usually in dB)

Terminology II

Term	Meaning in this topic
PCM	Pulse Code Modulation: sampling + quantization + binary encoding of sample levels
Bit depth (n_b)	Number of bits used per sample in PCM
Companding	Compressor + expander method to realize non-uniform effective quantization
DPCM	Differential PCM: transmits quantized prediction error instead of absolute sample
ADPCM	Adaptive DPCM: changes predictor/step behavior based on signal dynamics
Delta Modulation (DM)	1-bit differential signaling: indicate only up or down movement each sample
Slope overload	DM error when staircase cannot follow fast input changes
Granular noise	DM error when step size is too coarse for slowly varying input
ADM	Adaptive Delta Modulation: dynamically adjusts step size to reduce both DM error modes

Source Map I

Source anchors below follow the same mapping style used in the Chapter 4 source deck.

Slide portion	Primary source	Source anchor
Learning objectives, motivation slide (why conversion), and Topic 4 overview	Forouzan + instructor synthesis	Forouzan Ch. 4 digital transmission / ADC context, with instructor bridge to Topic 2/3 system flow
PCM encoder structure; sampling fundamentals; Nyquist condition and visual interpretation; Examples 4.1 and 4.2	Forouzan + Topic 2 base	Forouzan Ch. 4 analog-to-digital conversion, with Topic 2 Ch. 3 pp. 81–83 for Nyquist/Shannon background
Uniform/non-uniform quantization fundamentals and Example 4.3	Forouzan	Forouzan Ch. 4 quantization, companding, and SQNR discussion in the PCM sequence

Source Map II

Slide portion	Primary source	Source anchor
PCM encoding relation and Example 4.4; decoder and recovery intuition	Forouzan	Forouzan Ch. 4 PCM sequence (sampling, quantization, encoding, decoding)
PCM bandwidth relation and Example 4.5	Forouzan + line-coding linkage	Forouzan Ch. 4 PCM bit-rate discussion, connected to the Ch. 4 line-coding bandwidth relation
Why go beyond PCM; DPCM/ADPCM concept slides, worked examples, and comparison slide	Instructor synthesis + Forouzan context	Forouzan Ch. 4 PCM/DM differential-coding context; DPCM/ADPCM examples and comparison framing are instructor-generated

Source Map III

Slide portion	Primary source	Source anchor
Delta modulation slides: DM concept, modulator/demodulator, and 1-bit signaling block logic	Forouzan	Forouzan Ch. 4 delta modulation introduction and system view, corresponding to Ch. 4 figures 4.28–4.30 in the source deck
Two classical DM errors and adaptive delta modulation control logic	Forouzan + instructor redrawn figures	Forouzan Ch. 4 slope-overload/granular-noise and adaptive step-size discussion, adapted with course-specific explanatory diagrams
Applications, compression bridge, common confusions, summary, and terminology appendix	Instructor synthesis + Forouzan/Goldsmith context	Forouzan Ch. 4 ADC methods for the recap; Goldsmith Ch. 5 pp. 126–132 only for the digital-passband bridge to Topic 5